

Week 4 teaching guide

Make controls reliable · GROW computing · 40 minutes

I can create and test all four arrow controls in my maze.

Before the lesson

Open your own W03_Maze.sb3. Open the separate teacher model and check your school save route. Choose pupil booklets by current need. Keep the HTML or slides and the reference page ready.

Teacher example: W04_Teacher_Model.sb3. Pupil access file: W04_Recovery_No_Controls.sb3. Recovery supplies prior work only. Use the pupil's own project first.

Stage and time	Teach and check
Arrival 0-3	Open your own W03_Maze.sb3. Point to its safe starting place.
Starter 3-6	The up key changes x by 10. Which block would you inspect?
I Do 6-11	Right and left use x. Set a clear green-flag start.
We Do 11-18	Choose the correct axis for up. Reset before each new test.
I Do 2 18-21	Up and down use y. Use change y in each script.
We Do 2 · Task 21-25	Reset Player. Test and check the final position.
Independent Task 25-35	Build all four key scripts on Player. Explain one block and its effect.
Exit 35-40	Save as W04_Controls.sb3. Show which support or idea helped.

Independent Task: allow 25-32 for making and testing, then 32-35 for an individual demonstration and pupil voice. Exit stays at 35-40.

Choose the task route

Step by step: Build and test one key at a time.

Main task: Build four keys and test a return to the start.

Challenge: Test whether sprite direction changes x/y controls.

Teaching notes and answers

Use the teacher model for demonstration, then return to pupil construction

Use the pupil's own Week 3 maze. A recovery maze is supplied with no scripts; it supports access but cannot retrospectively prove creation of its sprites. Select Player before building. Model Events > when space key pressed and use the dropdown for each arrow. Use change x/change y, not set x/set y. Record the pupil's actual safe start; (-180,-120) is the supplied model example only. Keep steps at 10 for the shared tests. Test each key after reset using press and release. All four controls remain the target for every route; two keys are partial progress.

Block and design reference

Find this	Use it like this
Right arrow	Events: when right arrow key pressed. Motion: change x by 10.
Left arrow	Events: when left arrow key pressed. Motion: change x by -10.
Up arrow	Events: when up arrow key pressed. Motion: change y by 10.
Down arrow	Events: when down arrow key pressed. Motion: change y by -10.
Green flag	Events: when green flag clicked. Motion: go to x [start x] y [start y].

Expected answers

1. The broken up script changes x. Replace it with change y by 10. Changing only the sign still uses the wrong axis.
2. From the model start (-180,-120), single-key results are right (-170,-120), left (-190,-120), up (-180,-110), down (-180,-130). Reset before each.
3. Right, up, left, down returns to the start when all four step sizes are 10.
4. With direction 0, change x by 10 still goes right. Move 10 steps goes up. x/y controls work independently of facing direction.
5. Moving through a wall is expected at this stage. The collision consequence is the Week 5 task.

A likely misconception

Up uses positive y in Scratch. A negative change is a direction, not an error. The control script belongs on Player.

Access and evidence

Record the pupil action and the support used

Flexible support

Current need	Adaptation
Reading or memory	Use short instructions, the reference and one visible next action. Read aloud privately.
Communication	Accept speech, pointing, drawing or exact-word scribing. Give thinking time.
Keyboard or pointer access	Use an agreed school access device or let the pupil direct an adult. Record the method and check evidence arrangements with the UAS lead.
Regulation or attention	Offer quiet work, a pause and the same return step. Do not make a public performance necessary.
Ready to transfer	Use Challenge after the common goal. Ask for a prediction and explanation rather than extra writing.

Pupil voice with a response

Which key or action needs clearer help? Choose a useful support or step-size change and test it.

Ask “What I said, and what it changed”. Listen and reply: use it now, adapt it together, or defer it with a reason. Make any agreed change visible. An adult may record exact pupil words. No signature is required from the pupil’s audience.

Individual evidence

AQA 71638 outcome 2: Observe pupil creation of all four arrow scripts and a practical demonstration. Check any previously inherited controls by having the pupil construct the relevant script.

Pupil action or check	Date and support used
I created four separate key scripts on Player.	
Right/left change x; up/down change y.	
My reset uses a clear starting position.	
I saved, reopened and tested my controls.	

Record completed outcomes on the AQA summary sheet. Do not infer creation from supplied code. Outcomes 5 and 6 remain in Weeks 7 and 8; check every outcome individually.

If a device is unavailable

Use the four key/action cards. A partner gives the key input and the pupil moves a token on the grid. Return to building, running and saving all four scripts in Scratch.

[AQA 71638 Programming with Scratch unit 6](#)